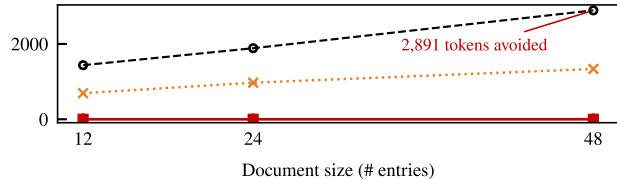


■ AgentTX causal (ours)    ✕ optimistic checkpoint    -●- whole-branch abort

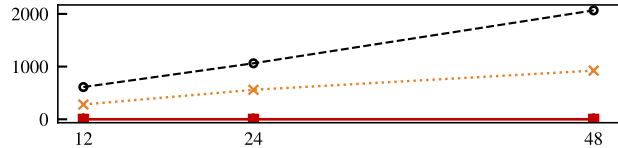
Replay tokens



Document size (# entries)

(a) Total LLM work

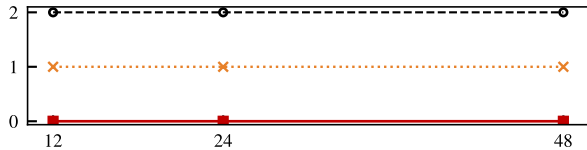
Completion tokens



Document size (# entries)

(b) Regenerated content

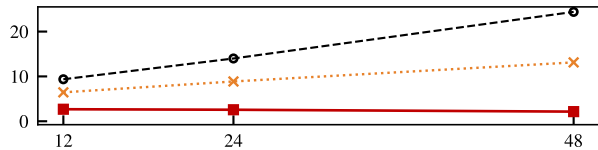
Documents regenerated



Document size (# entries)

(c) Valid work lost

Recovery p95 (s)



Document size (# entries)

(d) Replay latency